Ioannis Mandralis

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mandralis.github.io

Research

Autonomy for multimodal robotic systems that move and interact across aerial, terrestrial and aquatic environments. My work combines learning-based and model-predictive control with novel mechanical design, with particular emphasis on aerial platforms whose unconventional morphologies enable rich physical interaction, and on reinforcement learning for locomotion in fluid media.

Education

2021 – 10/2025 **California Institute of Technology,** Pasadena, USA
Ph.D. in Aeronautics, GPA 4.0/4.0
Thesis: "Leveraging Aerial Transformation for Enhanced Air–Ground Robotic Mobility"
Supervisors: Prof. Morteza Gharib and Prof. Richard M. Murray

2018 – 2020 **ETH Zürich,** Zürich, Switzerland
M.Sc. in Mechanical Engineering, specialisation in Robotics, GPA 5.85/6.0
Awarded the Degree Distinction Prize. Supervisor: Prof. Petros Koumoutsakos

2015 – 2018 **École Polytechnique Fédérale de Lausanne,** Lausanne, Switzerland
B.Sc. in Mechanical Engineering, GPA 5.68/6.0
One year of exchange study at McGill University, Montreal, Canada, GPA 4.0/4.0

Work

since 01/2026 **École Polytechnique Fédérale de Lausanne/Empa, Switzerland**
Postdoctoral Researcher with Prof. Mirko Kovač
Multimodal robotics: learning and control of robots operating across aerial, terrestrial and aquatic media

2021 –2026 **California Institute of Technology, Pasadena, USA**
Doctoral Researcher with Prof. Morteza Gharib and Prof. Richard M. Murray
Center for Autonomous Systems and Technologies (CAST)

2020 – 2021 **ETH Zürich**
Research Affiliate with Prof. Petros Koumoutsakos after end of M.Sc. (6 months)
Reinforcement learning for the control of swimming bodies in simulated flows

2019 – 2020 **ANYbotics AG, Zürich, Switzerland**
Robotics Internship during M.Sc. (6 months)
Development of autonomy stack for the ANYmal quadruped

Honours and Awards

2026 Hans G. Hornung Award, Caltech, for the best doctoral thesis defense presentation by a student advised by aerospace faculty.

2022 Onassis Foundation Scholarship, awarded in support of doctoral research in robotics.

2022 Rolf D. Buehler Memorial Prize in Aeronautics, awarded to the student ranked first in the Aeronautics Department.

2021 Caltech Doctoral Fellowship, full tuition and stipend for the duration of doctoral studies.

2020 Degree Distinction Prize, ETH Zürich, the highest degree distinction conferred by ETH Zürich.

2017 Scholarship for EPFL–McGill Mobility, awarded on the basis of academic achievement.

Journal Articles

[J1] I. Mandralis, R. Nemovi, A. Ramezani, R. M. Murray, and M. Gharib. ATMO: an aerially transforming morphobot for dynamic ground–aerial transition. *Nature Communications Engineering*, 2025. *link*

[J2] V. Gherold, I. Mandralis, E. Sihite, A. Salagame, A. Ramezani, and M. Gharib. Self-supervised cost of transport estimation for multimodal path planning. *IEEE Robotics and Automation Letters*, 10(7):6872–6879, 2025. *link*

[J3] I. Mandralis, S. Schumacher, and M. Gharib. Wake vectoring for efficient morphing flight. *(under review)*, 2026. *link*

[J4] A. A. Stefan-Zavala, I. Scherl, I. Mandralis, S. L. Brunton, and M. Gharib. Data-driven modelling for on-demand flow prescription in fan-array wind tunnels. *Flow*, Cambridge University Press, 2025. *link*

[J5] P. Gunnarson, I. Mandralis, G. Novati, P. Koumoutsakos, and J. O. Dabiri. Learning efficient navigation in vortical flow fields. *Nature Communications*, 12:7143, 2021. *link*

[J6] I. Mandralis, P. Weber, G. Novati, and P. Koumoutsakos. Learning swimming escape patterns under energy constraints. *Physical Review Fluids*, 6:093101, 2021. *link*

Conference Articles

[C1] I. Mandralis, R. M. Murray, and M. Gharib. Quadrotor morpho-transition: learning vs model-based control strategies. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Hangzhou, 2025. *link*

[C2] B. Gupta, A. Dhole, A. Salagame, X. Niu, Y. Xu, K. Venkatesh, P. Ghanem, I. Mandralis, E. Sihite, and A. Ramezani. Bounding flight control of dynamic morphing wings. In *IEEE International Conference on Advanced Intelligent Mechatronics (AIM)*, Boston, 2024. *link*

[C3] A. Salagame, S. Pandya, I. Mandralis, E. Sihite, A. Ramezani, and M. Gharib. NMPC-based unified posture manipulation and thrust vectoring for fault recovery. *(under review)*, 2025. *link*

[C4] A. Salagame, N. Bhattachan, A. Caetano, I. McCarthy, H. Noyes, B. Petersen, A. Qiu, M. Schroeter, N. Smithwick, K. Sroka, J. Widjaja, Y. Bohra, K. Venkatesh, K. Gangaraju, P. Ghanem, I. Mandralis, E. Sihite, A. Kalantari, and A. Ramezani. How strong a kick should be to topple Northeastern's tumbling robot. In *IEEE International Conference on Advanced Intelligent Mechatronics (AIM)*, Boston, 2024. *link*

[C5] I. Mandralis, E. Sihite, A. Ramezani, and M. Gharib. Minimum time trajectory generation for bounding flight: combining posture control and thrust vectoring. In *European Control Conference (ECC)*, Bucharest, 2023. *link*

[C6] E. Sihite, F. Slezak, I. Mandralis, A. Salagame, M. Ramezani, A. Kalantari, A. Ramezani, and M. Gharib. Demonstrating autonomous 3D path planning on a novel scalable UGV–UAV morphing robot. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Detroit, 2023. *link*

[C7] A. Dhole, B. Gupta, A. Salagame, X. Niu, Y. Xu, K. Venkatesh, P. Ghanem, I. Mandralis, E. Sihite, and A. Ramezani. Hovering control of flapping wings in tandem with multi-rotors. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Detroit, 2023. *link*

Patents

[T1] A. Rosakis, I. Mandralis, and M. Gharib. Ultrasound shape sensing. *US patent application US20250315966A1*, pending. *link*

[T2] I. Mandralis, S. Schumacher, and M. Gharib. Thrust recapture for morphing aerial vehicles with out-of-plane thrusters. *Patent disclosure filed*, 2024.

Talks

02/26 **Invited Seminars**
- McGill University (ME department)
- University of Wisconsin–Madison (ME department)
- Vanderbilt University (ME department)

10/25 **Quadrotor Morpho-Transition: Learning vs Model-Based Control Strategies**
- IROS, Hangzhou, China (oral presentation)

04/25 **Ground–Aerial Multimodal Robotics: The Aerially Transforming Morphobot**
- HiPeR Lab, University of California, Berkeley (invited talk)

10/24 **Multimodal Robotics Booth with the Technology Innovation Institute**
- IROS, Abu Dhabi, United Arab Emirates

06/23 **Minimum Time Trajectory Generation for Bounding Flight**
- European Control Conference, Bucharest, Romania

11/23 **Thrust Recapture for Morphing Aerial Vehicles with Out-of-Plane Thrusters**
- APS Division of Fluid Dynamics, Washington DC, USA

11/21 **Investigation of Surface Pressure Fluctuations in Airfoils: Towards Gust Mitigation**
- APS Division of Fluid Dynamics, Phoenix, USA

Teaching

Caltech **Teaching Assistant,** Pasadena, USA
Courses:
- Optimal Control & Estimation (CDS 112 / Ae 103b), Winter 2023, Prof. Richard M. Murray
- Linear Systems Theory, Fall 2023, Prof. Amir Rahmani

ETH Zürich **Teaching Assistant,** Zürich, Switzerland
Course:
- Multivariable Control, Spring 2020, Prof. Roy Smith

EPFL **Teaching Assistant,** Lausanne, Switzerland
Courses:
- Thermodynamics, Spring 2017, Prof. Véronique Michaud
- Analytical Mechanics, Fall 2016, Prof. Philippe Müllhaupt

Students Supervised

Home institution of visiting students given in parentheses.

EPFL **Doctoral**
Junghwan Ro (2026–): aerial–aquatic escape and medium transition using soft materials and reinforcement-learning based control

Master's theses
Alexandros Pantalos (2026–): vision–language–action models for robotic navigation and exploration trained on synthetic-only Gaussian splatting data

Caltech **Master's theses**
Simon Gmur (EPFL, 2025), Quentin Delfosse (EPFL, 2024), Gabriel Margaria (EPFL, 2024), Vincent Gherold (EPFL, 2024, Prix BlueBotics for the best Master's thesis in robotics at EPFL), Severin Schumacher (TU Munich, 2023)

Undergraduate research
Diego Garcia (Caltech, 2024), Sofia Wan & Steven Lei (Caltech, 2023, finalist best SURF project)

Academic Services

Reviewer
- IEEE Transactions on Robotics (T-RO)
- IEEE Robotics and Automation Letters (RA-L)
- Nature Scientific Reports
- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- European Control Conference (ECC)

Outreach
- Led more than fifteen tours of the Caltech Center for Autonomous Systems and Technologies for investors, media, student groups and schools

Media Coverage

2025 A Symphony of Robotic Motion — Caltech and the Technology Innovation Institute
2025 Mid-Air Transformation Helps Flying, Rolling Robot to Transition Smoothly — Caltech News
2024 Sky News Arabia coverage of ATMO at IROS 2024
2021 Engineers Teach AI to Navigate the Ocean with Minimal Energy — Caltech News

References

Morteza Gharib
Hans W. Liepmann Professor of Aeronautics and Medical Engineering; Director of GALCIT and CAST, California Institute of Technology
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Richard M. Murray
Thomas E. and Doris Everhart Professor of Control and Dynamical Systems and Bioengineering, California Institute of Technology
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John O. Dabiri
Centennial Professor of Aeronautics and Mechanical Engineering, California Institute of Technology
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